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CW-VAE: Shape Generation via Conformal Welding Priors in Variational Autoencoders*

Chenran Lin
Oscar Yau Lam Chau
and Lok Ming Lui

Generating shapes remains challenging in computer vision and graphics, particularly with limited training data. Unlike general image generation, shape-focused approaches provide advantages in geometric validity, data efficiency, and interpretability for tasks like segmentation and registration. This paper presents CW-VAE (Conformal Welding Variational Autoencoder), a framework using conformal welding as a geometric prior for 2D simply-connected shapes. Our method extracts boundary points, computes their conformal welding representations, and trains a specialized VAE with a composite loss function to ensure geometric validity. Experiments on animal silhouettes and medical lung contours show that CW-VAE produces diverse, high-quality shapes even with limited training data, outperforming methods using boundary points or pixel representations directly. This work demonstrates how geometric priors enhance generative models for shape synthesis in data-scarce scenarios.

AMS 2000 subject classifications: shape generation, conformal welding, variational autoencoder, deep generative models.

1. Introduction

Shape generation is fundamental to computer graphics, computer vision, and related fields, with applications ranging from entertainment content creation to engineering design augmentation and medical imaging visualization. Recent machine learning advances have enabled significant progress in synthesizing novel and realistic data, including images, text, and audio. Models like Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and diffusion models demonstrate strong capabilities in learning complex data distributions and generating high-fidelity samples.

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Despite these developments, creating diverse and high-quality shapes from limited datasets remains challenging. Many generative models require substantial training data for adequate performance, with effectiveness diminishing when data is scarce. This limitation is particularly relevant in specialized domains where collecting large datasets is often impractical. Furthermore, objects of interest frequently exhibit distinct geometric features and structural properties—shape priors—that generic generative models may not effectively capture or utilize, especially with sparse data. Integrating such priors offers a promising approach to enhance sample quality, ensure geometric validity, and improve learning efficiency.

Focusing specifically on shape generation rather than general image synthesis provides several advantages. Shape representations offer stronger geometric guarantees and topological correctness, crucial for applications like medical image analysis where anatomical validity is essential. Shape-centric models typically require less training data and perform better in small-sample scenarios by leveraging geometric priors. Additionally, shape representations are more interpretable and controllable, facilitating integration with traditional algorithms for tasks like segmentation, registration, and shape matching. This makes shape-first approaches particularly valuable where geometric accuracy outweighs texture details.

This paper addresses shape generation from small datasets by proposing a novel method integrating a powerful geometric shape prior: *conformal welding* [22]. We introduce the Conformal Welding Variational Autoencoder (CW-VAE), designed to learn and generate 2D simply-connected shapes. Conformal welding provides a concise representation of a simply-connected shape through a unit circle homeomorphism, effectively capturing intrinsic geometric information.

The CW-VAE framework operates as follows: For each shape in a class, we extract boundary points and compute the corresponding conformal welding function. These representations train the CW-VAE, which learns a latent distribution over conformal weldings. New shapes are generated by sampling this latent space to produce novel conformal welding functions, which are then reconstructed into 2D shapes.

The main contributions of this work are:

- We propose a novel generative model, CW-VAE, that effectively integrates conformal welding as a shape prior for generating 2D simply-connected shapes, demonstrating strong performance even with limited training data.

- We develop a pipeline with carefully designed pre- and post-processing steps that significantly improve the VAE's ability to learn and generate conformal weldings, including normalization techniques to make conformal weldings more amenable to VAE learning, and efficient reconstruction methods to transform VAE outputs back into shapes.
- We empirically validate our approach through experiments, showcasing its ability to generate high-quality and diverse shapes and highlighting the benefits of incorporating conformal welding priors compared to baseline methods.

This paper is organized as follows[17]. Section 2 reviews related work in shape generation and the use of shape priors. Section 3 provides necessary background on quasi-conformal mappings, conformal welding, and Variational Autoencoders. Section 4 details our proposed CW-VAE framework, including data preprocessing, the VAE architecture, postprocessing, and the loss function design. Section 5 presents the experimental setup, datasets, evaluation metrics, and a comprehensive analysis of the results. Finally, Section 6 concludes the paper and discusses potential directions for future research.

2. Related Work

2.1. Shape Priors in Computer Vision and Graphics

The concept of a shape prior is fundamental in computer vision, graphics, and medical image analysis, referring to explicit or implicit constraints on the geometry or topology of objects. Incorporating prior knowledge about expected shapes allows algorithms to overcome challenges posed by noise, occlusion, limited data, or ambiguity in visual information. Shape priors guide models towards plausible and regular solutions, significantly improving performance in tasks such as segmentation [7], registration [27], object recognition [9], and 3D reconstruction [3].

A wide variety of shape representations have been developed to encode prior knowledge. Early approaches often relied on statistical shape models (SSMs) like Active Shape Models (ASMs) and Active Appearance Models (AAMs) [6], which learn a principal component analysis (PCA) based model of shape variation from a training set of landmark-aligned examples. Template-based methods utilize one or more exemplary shapes that are deformed to fit new instances. Part-based models, such as pictorial structures [8], represent shapes as a collection of parts arranged in a deformable configuration.

Other representations focus on capturing intrinsic geometric properties. Boundary-based descriptors like Fourier Descriptors [25] and wavelet descriptors capture the contour information. Region-based methods such as Zernike moments or the Medial Axis Transform (MAT) [4] describe the interior and skeletal structure of a shape. More recently, learning-based approaches, particularly those using deep neural networks, have enabled the development of powerful implicit shape priors by learning low-dimensional embeddings of shapes directly from data [21]. These learned priors can capture complex shape variations and have become central to many modern generative models for shapes. The choice of shape prior often depends on the specific application, the nature of the shapes being modeled, and the desired trade-off between representational power and computational tractability. Our work explores the use of conformal welding as a specific, geometrically rich prior for 2D simply-connected shapes within a deep generative framework.

2.2. Quasi-conformal Theory and Shape Representation

Quasi-conformal theory has emerged as a powerful mathematical framework for shape analysis and representation in computer vision, offering unique advantages for capturing geometric deformations while preserving topological properties [24, 18, 12]. The theory's ability to represent complex shape transformations through Beltrami coefficients has made it particularly valuable for shape generation and manipulation tasks.

A significant advancement was introduced by Chan et al. [5], who demonstrated how the Beltrami representation of shapes can be leveraged for topology-preserving applications. Building upon this, Zhang et al. [26] combined quasi-conformal theory with deep learning, introducing a Topology-Preserving Segmentation Network (TPSN) that demonstrated robust performance in medical imaging applications.

The success of these quasi-conformal frameworks motivates our use of conformal welding as a shape prior in the CW-VAE. Our work extends this line of research by demonstrating how conformal welding can be effectively integrated into deep generative models for shape synthesis. The high-quality shapes generated by CW-VAE can serve as valuable training data for downstream tasks such as segmentation, registration, and other computer vision applications where geometric consistency is crucial.

2.3. Variational Autoencoders (VAEs) for Shape Generation

Variational Autoencoders (VAEs) [16], as detailed in Section 3.3, provide another prominent framework for generative modeling. VAEs learn a probabilistic mapping from input data to a lower-dimensional latent space and a generative mapping from the latent space back to the data space. They are trained by maximizing the Evidence Lower Bound (ELBO), which balances reconstruction accuracy with a regularization term (KL divergence) that encourages the latent space to follow a prior distribution, typically a Gaussian. This regularization facilitates a smooth and structured latent space, making VAEs suitable for tasks like interpolation and controlled generation.

In the context of shape generation, VAEs have been applied to various representations. For 2D shapes, VAEs can learn to generate boundary contours or implicit representations. For 3D shapes, VAEs have been adapted to process point clouds, volumetric data, and mesh-based representations. For example, architectures similar to those used for point cloud processing with GANs, such as those incorporating PointNet-like encoders, have also been employed within VAE frameworks [1]. VAEs have also been used to learn generative models of Signed Distance Functions (SDFs) or other implicit shape representations [21, 20], where the VAE learns a distribution over functions that define shapes.

Compared to GANs, VAEs generally offer more stable training dynamics and a more interpretable latent space. While VAEs traditionally have been associated with generating somewhat blurrier samples than GANs in image generation, this characteristic is less of a direct concern when the output is a structured shape representation (e.g., a set of parameters or a discrete boundary) rather than a pixel grid. However, challenges remain in ensuring that VAEs capture the full complexity and diversity of shape distributions, especially for intricate 3D objects. The design of the VAE architecture, particularly the decoder, and the choice of reconstruction loss are critical for generating high-fidelity and geometrically valid shapes. Our proposed CW-VAE builds upon this VAE framework, specifically tailoring it to the conformal welding representation to leverage its geometric properties.

2.4. Generative Models Incorporating Geometric Priors

While general deep generative models can learn complex shape distributions from raw data (e.g., point clouds or voxels), their performance can be enhanced, particularly in data-scarce regimes or when precise geometric control is desired, by explicitly incorporating domain-specific shape priors. This involves designing generative frameworks that operate on or are guided by more structured geometric representations rather than raw, unstructured data.

Several approaches have explored integrating various forms of geometric priors into generative models. For instance, some works have focused on generating parameters of traditional shape models, such as statistical shape models (SSMs) or CAD-based parametric models, using GANs or VAEs. This allows the generative network to operate in a lower-dimensional, often more interpretable, parameter space where geometric validity can be more easily enforced [10]. Other methods have sought to combine learned implicit representations with explicit geometric constraints. For example, generative models for signed distance functions (SDFs) might incorporate priors that encourage smooth surfaces or specific topological characteristics [21].

The motivation for integrating such priors is manifold: it can lead to more robust learning from smaller datasets, as the model does not need to learn fundamental geometric principles from scratch; it can provide better control over the generation process by manipulating meaningful geometric parameters; and it can improve the quality and plausibility of the generated shapes by ensuring they adhere to known constraints.

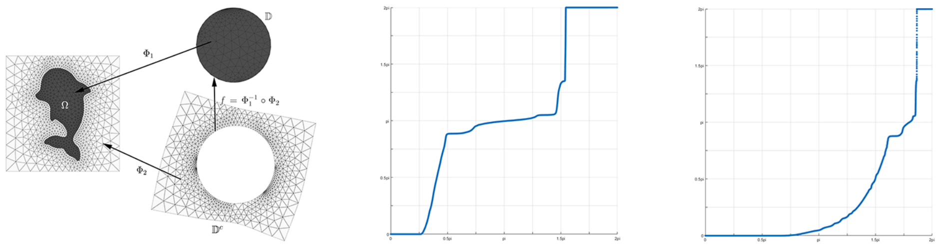
Our proposed Conformal Welding Variational Autoencoder (CW-VAE) aligns with this direction of research. By employing conformal welding as the underlying shape representation, we leverage a powerful and canonical geometric prior for 2D simply-connected shapes. Conformal welding encapsulates the intrinsic geometry of a shape via a homeomorphism of the unit circle, offering a compact and theoretically well-founded descriptor. Training a VAE directly on these conformal welding functions, as opposed to raw boundary points or pixel-based representations, allows the model to learn in a space where geometric properties are more explicitly encoded. This approach aims to improve the quality and diversity of generated shapes, especially when training data is limited, and to ensure the generation of geometrically valid and meaningful forms. The novelty of CW-VAE lies in this specific combination of a classical geometric tool with a modern deep generative framework for the task of 2D shape synthesis.

3. Background

3.1. Conformal Welding

Conformal welding provides a foundational method for constructing and characterizing a simply-connected Jordan domain $\Omega \subset \mathbb{C}$. The technique involves welding together the boundary values of two conformal maps: one maps the unit disk, $\mathbb{D}$, to Ω, and another maps the exterior of the unit disk, $\mathbb{D}^c$, to the complement of Ω, denoted as Ω^c.

Formally, consider a simply-connected Jordan domain Ω. The Riemann mapping theorem guarantees the existence of a conformal map $\Phi_1 : \mathbb{D} \to \Omega$ and another conformal map $\Phi_2 : \mathbb{D}^c \to \Omega^c$. Both Φ_1 and Φ_2 can be extended



(a) (b) (c)

Figure 1: (a) the illustration of the method to construct conformal welding from a given simply-connected shape; (b), (c) shows 2 different conformal welding functions H of the shape in (a).

to homeomorphisms from their respective closed domains (i.e., $\overline{\mathbb{D}}$ and $\overline{\mathbb{D}^c}$) to the closures $\overline{\Omega}$ and $\overline{\Omega^c}$. The conformal welding of Ω is then defined as the orientation-preserving homeomorphism $h : \partial\mathbb{D} \to \partial\mathbb{D}$ obtained by composing these boundary maps:

$$h = \Phi_1^{-1} \circ \Phi_2. \tag{1}$$

Conversely, any orientation-preserving homeomorphism $h : \partial\mathbb{D} \to \partial\mathbb{D}$ qualifies as a conformal welding if it can be extended to a quasi-conformal map on $\mathbb{C}$ while it remains conformal in $\mathbb{D}^c$.

The welding homeomorphism h can be conveniently represented by a monotonically increasing function $H : [0, 2\pi] \to [0, 2\pi]$, where $h(e^{i\theta}) = e^{iH(\theta)}$. This function H must satisfy $H(0) = 0$ and $H(2\pi) = 2\pi$, effectively acting as a reparameterization of the unit circle's boundary.

Despite its theoretical elegance, the conformal welding h (or its functional form H) obtained for a given shape Ω is not inherently unique, Figure 1 shows 2 examples of the conformal welding function H for the same shape. This non-uniqueness arises because the conformal maps Φ_1 and Φ_2 are themselves unique only up to Möbius transformations of their respective domains ($\mathbb{D}$ and $\mathbb{D}^c$ respectively). Different choices of these normalizing Möbius transformations will yield different, albeit related, welding homeomorphisms for the identical shape Ω. Consequently, for practical use in shape analysis and generative modeling, it is crucial to apply a normalization procedure to the conformal welding. This ensures that each distinct shape maps to a unique representative welding function.

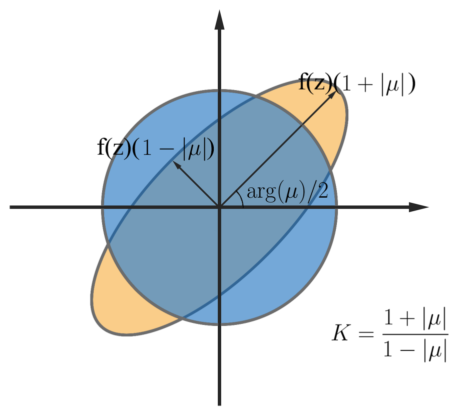


Figure 2: Illustration of the quasi-conformal mapping: quasi-conformal maps transform infinitesimal circles into ellipses. The Beltrami coefficient quantifies the distortion or dilation of these ellipses under the quasi-conformal mapping.

3.2. Quasi-conformal Mappings

A complex function $f : \Omega \subset \mathbb{C} \to \mathbb{C}$ is characterized as *quasi-conformal* [2, 11] if it preserves orientation and adheres to the Beltrami equation:

$$\frac{\partial f}{\partial \overline{z}} = \mu(z)\frac{\partial f}{\partial z} \tag{2}$$

In this equation, $\mu(z)$ represents a complex-valued Lebesgue measurable function, termed the Beltrami coefficient of f, which must satisfy the condition $\|\mu\|_\infty < 1$. This coefficient $\mu : \Omega \to \mathbb{D}$ is explicitly defined as:

$$\mu = \frac{f_{\overline{z}}}{f_z} \tag{3}$$

The geometric impact of f can be understood by examining the pullback of the Euclidean metric ds_E^2. This transformed metric under f is expressed as:

$$f^*(ds_E^2) = \left|\frac{\partial f}{\partial z}\right|^2 |dz + \mu(z)d\overline{z}|^2 \tag{4}$$

When analyzed with respect to the standard Euclidean metric components dz and $d\overline{z}$, this pullback metric exhibits eigenvalues $(1+|\mu|)^2\left|\frac{\partial f}{\partial z}\right|^2$ and $(1-$

$|\mu|)^2 \left|\frac{\partial f}{\partial z}\right|^2$. These eigenvalues directly relate to the principal stretches of the mapping.

Locally, around a point $p \in \Omega$, the mapping f can be decomposed. It can be viewed as a translation to $f(p)$ followed by the action of a stretch map $S(z) = z + \mu(p)\overline{z}$ and multiplication by the complex derivative $f_z(p)$. This local behavior is captured by:

$$f(z) = f(p) + S(z)f_z(p) = f(p) + (z + \mu(p)\overline{z})f_z(p). \tag{5}$$

The term $S(z)$ is responsible for transforming an infinitesimal circle into an infinitesimal ellipse, with the Beltrami coefficient μ entirely dictating this local geometric distortion. The properties of $\mu(p)$ determine the characteristics of this ellipse: the direction of maximum magnification is given by $\arg(\mu(p))/2$, with a corresponding magnification factor of $1 + |\mu(p)|$. Orthogonally, the direction of maximum shrinkage is $\arg(\mu(p))/2 - \pi/2$, with a shrinkage factor of $1 - |\mu(p)|$. The ratio of these factors defines the local distortion or dilation $K(p)$:

$$K(p) = \frac{1 + |\mu(p)|}{1 - |\mu(p)|}.$$

Consequently, the Beltrami coefficient μ provides comprehensive information about the mapping's local geometric properties and serves as a direct measure of its deviation from conformality. Specifically, f is conformal in a neighborhood of p if $\mu(p) = 0$. If $\mu(z) = 0$ for all $z \in \Omega$, then f is conformal (or holomorphic) throughout Ω.

A crucial aspect of quasi-conformal theory is the direct correspondence between a quasi-conformal map f and its Beltrami coefficient μ. As defined in equation 3, a given f uniquely determines μ. Conversely, a fundamental theorem (the Measurable Riemann Mapping Theorem) establishes that for any admissible Beltrami coefficient μ (i.e., measurable with $\|\mu\|_\infty < 1$), there exists a quasi-conformal map f whose Beltrami coefficient is μ.

3.3. Variational Autoencoders (VAEs)

Variational Autoencoders (VAEs), pioneered by Kingma and Welling, constitute a significant category of deep generative models. They are designed to learn a compressed representation of high-dimensional data within a lower-dimensional latent space by synergizing principles from variational Bayesian methods and neural networks. The appeal of VAEs lies in their capacity for generating novel data samples and their ability to learn a well-structured, often continuous, latent space that supports meaningful interpolation and exploration.

Architecturally, a VAE comprises two primary neural network components:

- **Encoder (Inference Network):** This network, commonly denoted $q_\phi(z|x)$, processes an input data sample x and outputs the parameters (typically the mean μ_z and variance σ_z^2) of a probability distribution (e.g., a Gaussian $\mathcal{N}(\mu_z, \mathrm{diag}(\sigma_z^2))$) in the latent space. Rather than producing a deterministic latent vector, the encoder defines this distribution, from which a latent vector z is stochastically sampled. This sampling is a hallmark of VAEs.
- **Decoder (Generative Network):** This network, denoted $p_\theta(x|z)$, receives a latent vector z as input. During training, z is sampled from the encoder's output distribution $q_\phi(z|x)$; for generation, z is sampled from a chosen prior distribution, such as a standard normal $\mathcal{N}(0, I)$. The decoder then maps z back to the original data space, aiming to reconstruct x or generate a new sample x'. The decoder's output typically parameterizes a distribution over the data (e.g., Gaussian for continuous data, Bernoulli for binary data).

VAEs are trained by optimizing an objective related to the marginal likelihood $p(x)$ of the observed data. Since direct maximization of $p(x)$ is generally intractable, VAEs instead maximize a lower bound on its logarithm, known as the Evidence Lower Bound (ELBO):

$$\mathcal{L}_{\mathrm{ELBO}}(\phi, \theta; x) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - D_{\mathrm{KL}}(q_\phi(z|x) \| p(z)). \tag{6}$$

This objective function consists of two key terms:

- The first term, $\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)]$, is the expected reconstruction likelihood. It penalizes deviations between the original input x and its reconstruction from the latent sample z, effectively encouraging the decoder $p_\theta(x|z)$ to produce faithful reconstructions. This term is often realized as a mean squared error or a binary cross-entropy, depending on the data type.
- The second term, $D_{\mathrm{KL}}(q_\phi(z|x) \| p(z))$, represents the Kullback-Leibler (KL) divergence between the approximate posterior distribution $q_\phi(z|x)$ learned by the encoder and a pre-defined prior distribution $p(z)$ over the latent variables (commonly $\mathcal{N}(0, I)$). This KL divergence acts as a regularization term, compelling the encoded distributions to stay close to the prior, which promotes a more organized latent space and facilitates coherent generation from $p(z)$.

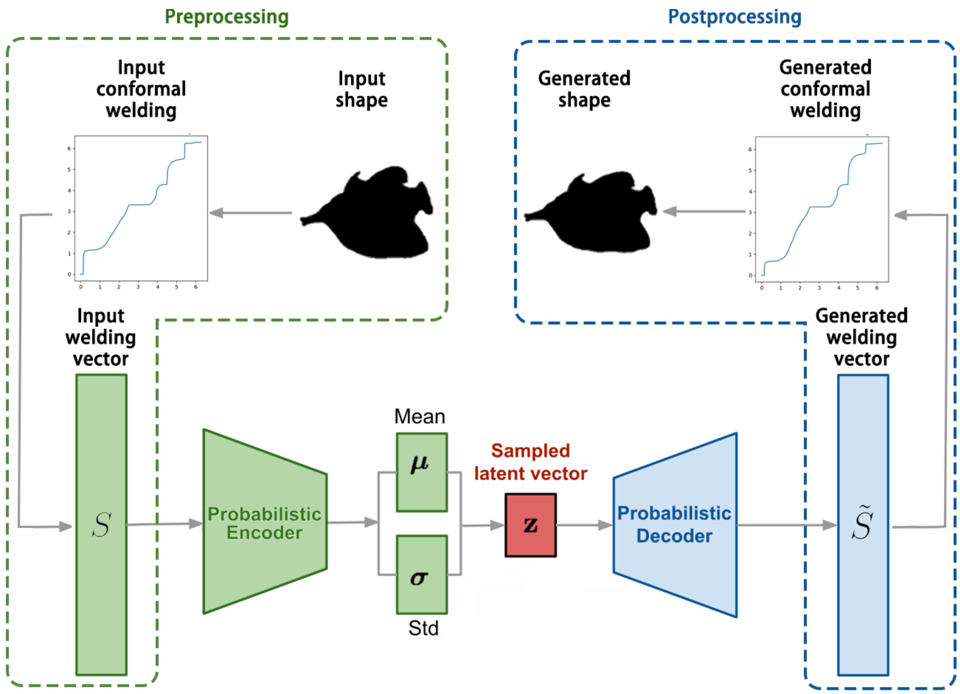


Figure 3: Architecture of the Conformal Welding Variational Autoencoder (CW-VAE).

Training typically involves minimizing the negative ELBO. A critical component for effective training is the reparameterization trick (e.g., for a Gaussian latent $z = \mu_z + \sigma_z \odot \epsilon$, with $\epsilon \sim \mathcal{N}(0, I)$), which allows gradients to be backpropagated through the stochastic sampling process.

Once trained, a VAE can generate new data by drawing samples z from the prior $p(z)$ and passing them through the decoder $p_\theta(x|z)$. VAEs are recognized for their training stability, especially compared to GANs, and their ability to learn smooth, semantically meaningful latent spaces. While they have sometimes been noted for producing slightly less sharp samples than GANs in image generation, their robust framework makes them suitable for a wide array of tasks, including image synthesis, anomaly detection, and representation learning.

4. Proposed Method

This section introduces the Conformal Welding Variational Autoencoder (CW-VAE), a framework for shape generation that incorporates conformal welding as a geometric prior for 2D simply-connected shapes. The approach involves three principal stages: (1) preprocessing and feature extraction from shapes, (2) generative modeling using a VAE architecture adapted for conformal welding representations, and (3) postprocessing to reconstruct new shapes from generated conformal weldings.

The key steps of our method are as follows:

- **Preprocessing:** For each given input shape Ω_k, we extract its boundary points and compute the corresponding conformal welding function H_k. To ensure uniqueness and facilitate learning, we normalize the conformal welding and apply a value re-distribution strategy to achieve welding vector S_k.
- **CW-VAE Generation:** The set of processed welding vector representations $\{S_k\}$ are used to train a VAE, which learns a latent distribution over the space of shape priors.
- **Postprocessing:** New welding vectors $\{\tilde{S}_l\}$ are generated by sampling from the latent space and decoding. They are mapped back to conformal welding functions $\{\tilde{H}_l\}$ and finally reconstructed as novel generated shapes $\{\tilde{\Omega}_l\}$.

In the following subsections, we detail each component of the framework.

4.1. Preprocessing

The CW-VAE model utilizes a processed conformal welding representation termed the *welding vector*. The preprocessing pipeline transforms the boundary of a 2D simply-connected shape Ω into a welding vector S through several stages.

4.1.1. Initial Conformal Welding Computation For each input shape Ω, an ordered sequence of M boundary points $\{p_j\}_{j=1}^{M}$ is extracted. The conformal welding $h : \partial\mathbb{D} \to \partial\mathbb{D}$ is computed based on Ω's geometry, following Section 3.1.

4.1.2. Normalization and Discretization of Conformal Welding As discussed in Section 3.1, the conformal welding function h exhibits non-uniqueness due to the conformal automorphisms of Riemann mappings. To establish a canonical representation conducive to learning, we employ normalization techniques analogous to those utilized in the normalization of Harmonic Beltrami Signatures (HBS) [17], which are given by the following conditions:

$$\begin{aligned} \int_{\partial\Omega} \Phi_1^{-1}(z)dz = 0, \quad \Phi_2(\infty) = \infty, \\ \arg\int_{\mathbb{D}} B(z)dz = 0, \quad \arg\int_{\mathbb{D}} \frac{B(z)}{z}dz \in [0,\pi), \end{aligned} \tag{7}$$

where B denotes the GHBS of h, ensure uniqueness as proven in [17]. Consequently, the corresponding conformal welding h becomes unique.

The normalized h is represented as a monotonically increasing function $H : [0, 2\pi] \to [0, 2\pi]$ satisfying $h(e^{i\theta}) = e^{iH(\theta)}$, with $H(0) = 0$ and $H(2\pi) = 2\pi$. Discretization samples $[0, 2\pi]$ at $N + 1$ equally spaced points, forming a discrete *function value vector*:

$$Y = \{y_i = H(\frac{2\pi i}{N})\}_{i=0}^{N}. \tag{8}$$

The *derivative vector* is computed from Y:

$$D = \{d_i = y_i - y_{i-1}\}_{i=1}^{N}. \tag{9}$$

Due to the monotonicity of H, we have $d_i > 0$ for all i, and $\sum_{i=1}^{N} d_i = y_N - y_0 = 2\pi$. This derivative vector D captures the local rate of change of the welding function.

4.1.3. Value Re-distribution An empirical analysis of the derivative vector D reveals that its distribution is often highly skewed and long-tailed, as shown in Figure 4. Typically, a large number of d_i values are concentrated in a small range slightly greater than zero, while a few values can be much larger. The strict positivity $d_i > 0$ also creates a sharp truncation at zero in the distribution. This characteristic can make it challenging for a standard VAE to effectively model the distribution, especially if the decoder of VAE uses common activation functions (e.g., sigmoid or tanh) that produce outputs in a bounded range or assume a more symmetric data distribution.

To address this and map the derivative vector D to a more favorable range for VAE learning, we apply a value re-distribution. Specifically, we transform each d_i into s_i using the exponential function to obtain the *welding vector* S:

$$S = \{s_i = e^{-d_i}\}_{i=1}^{N}. \tag{10}$$

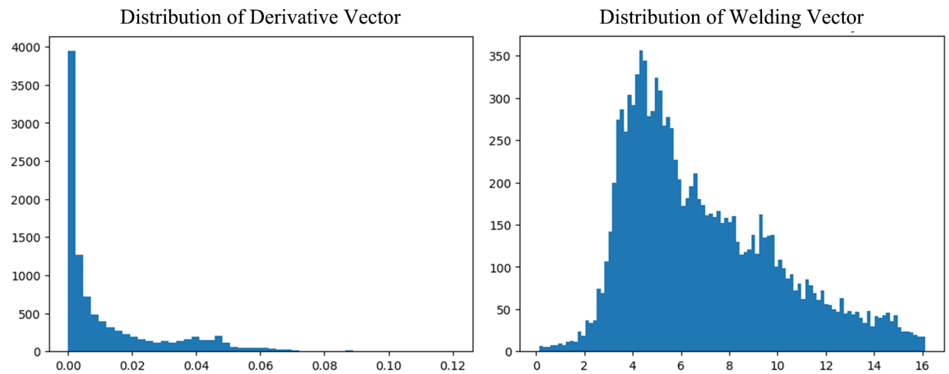


Figure 4: Left: the histogram of the value distribution of the derivative vector D; Right: the histogram of the value distribution of the welding vector S after re-distribution.

Since $d_i > 0$, the transformed values s_i will lie in the range $(0, 1)$. This transformation compresses larger d_i values towards 0 and maps smaller d_i values (those close to 0) to s_i values close to 1. This results in a more regular distribution, as shown on the right in Figure 4, mitigating issues related to skewed tails and improving the VAE's ability to model the data.

The welding vector S serves as the final feature vector representing shape Ω, providing input to CW-VAE for training and generation.

4.2. Model Architecture

The CW-VAE framework centers on learning the distribution of welding vectors S. Its architecture consists of two primary components: an encoder that maps input S to a latent space representation, and a decoder that reconstructs S from latent space samples. Both components employ neural networks with residual connections, drawing inspiration from ResNet [13] to enable deeper architectures while maintaining training stability. We implement these connections through Residual Linear Blocks.

4.2.1. Residual Linear Block A Residual Linear Block (ResLinearBlock) forms the fundamental component of our encoder and decoder networks. Each ResLinearBlock takes an input tensor and processes it through an input layer (fully connected linear layer followed by ReLU activation and Layer Normalization), a configurable number of repeated hidden layers (each also a linear layer followed by ReLU and Layer Normalization), and an output layer (again, linear, ReLU, and Layer Normalization). A shortcut connection is added from the input of the block to the output of these layers, followed by a final ReLU activation. If the input and output dimensions of the block differ, the shortcut connection involves a linear projection to match dimensions; otherwise, it is an identity mapping. This structure helps in mitigating the vanishing gradient problem and allows for the construction of deeper and more expressive networks.

4.2.2. Encoder The encoder $q_\phi(z|S)$ approximates the posterior distribution of latent variables z given welding vector S. A sequence of ResLinearBlocks progressively transforms S into higher-level representations. The final block's output feeds into two separate linear layers: one outputs mean vector μ_z, the other outputs log-variance $\log \sigma_z^2$. These parameters define the latent Gaussian distribution $q_\phi(z|S) = \mathcal{N}(z; \mu_z, \mathrm{diag}(\sigma_z^2))$, where input dimension N reduces to latent dimension L_{dim}.

A typical ResLinearBlock sequence in the encoder reduces dimensionality progressively (e.g., $N \to 512 \to 256 \to 128$). Separate linear layers then project the final 128-dimensional output to L_{dim}-dimensional μ_z and $\log \sigma_z^2$.

4.2.3. Decoder The decoder network, $p_\theta(\tilde{S}|z)$, takes a latent vector z (sampled from $q_\phi(z|S)$ during training or from the prior $p(z) = \mathcal{N}(0, I)$ during generation) and aims to reconstruct the original welding vector S. The architecture of the decoder mirrors that of the encoder, consisting of a sequence of ResLinearBlocks that transform the latent vector z back into the feature space. The output of the final ResLinearBlock is passed through a fully connected linear layer followed by a sigmoid activation function to produce the reconstructed welding vector $\tilde{S} = \{\tilde{s}_1, \ldots, \tilde{s}_N\}$. The sigmoid ensures that the output values $\tilde{s}_i$ lie in the range $(0, 1)$, consistent with the re-distribution applied during preprocessing.

4.3. Postprocessing

Once the CW-VAE has been trained, it can generate new welding vectors

$$\tilde{S} = \{\tilde{s}_1, \ldots, \tilde{s}_N\}$$

by sampling a latent vector z from the prior distribution $p(z) = \mathcal{N}(0, I)$ and passing it through the decoder network $p_\theta(\tilde{S}|z)$. To transform this generated welding vector $\tilde{S}$ back into a 2D shape, a series of postprocessing steps are applied, effectively reversing the preprocessing operations.

4.3.1. Inverse Value Re-distribution The first step recovers the derivative vector $\tilde{D}$ from the generated welding vector $\tilde{S}$ using:

$$\tilde{D} = \{\tilde{d}_i = -\log(\tilde{s}_i + \epsilon)\}_{i=1}^{N}. \tag{11}$$

Since the decoder outputs $\tilde{s}_i$ are expected to be in $(0, 1)$, $\tilde{d}_i$ will be positive. A small positive ϵ is added to $\tilde{s}_i$ to enhance numerical stability by preventing $\log(0)$ when $\tilde{s}_i$ approaches zero.

4.3.2. Accumulation and Scaling Next, we construct the unnormalized function value vector $\tilde{Y}' = \{\tilde{y}'_j\}_{j=0}^{N}$ through cumulative summation:

$$\tilde{y}'_0 = 0, \quad \tilde{y}'_j = \sum_{i=1}^{j} \tilde{d}_i \quad \text{for } j = 1, \ldots, N. \tag{12}$$

The sum $\tilde{y}'_N = \sum_{i=1}^{N} \tilde{d}_i$ may not exactly equal 2π. To enforce the fundamental property of the corresponding conformal welding function $\tilde{H}$, where $\tilde{H}(0) = 0$ and $\tilde{H}(2\pi) = 2\pi$, we apply a linear scaling to obtain the final function value vector $\tilde{Y}$:

$$\tilde{Y} = \{\tilde{y}_j = \tilde{y}'_j \cdot \frac{2\pi}{\tilde{y}'_N}\}_{j=0}^{N}. \tag{13}$$

If $\tilde{y}'_N$ is zero or very close to it (an unlikely event in a well-trained model), we can default to a uniform distribution, i.e., $\tilde{y}_j = \frac{2\pi j}{N}$, which corresponds to a circle.

The above steps ensure that the resulting function value vector $\tilde{Y}$ correctly spans the $[0, 2\pi]$ interval with $\tilde{y}_0 = 0$ and $\tilde{y}_N = 2\pi$, yielding a valid discrete representation of a normalized conformal welding function $\tilde{H}$.

4.3.3. Shape Reconstruction The final step reconstructs the 2D shape boundary from the discrete conformal welding function $\tilde{H}$ represented by $\tilde{Y}$. This requires solving the inverse problem: given welding homeomorphism $\tilde{h}(e^{i\theta}) = e^{i\tilde{H}(\theta)}$, find the Jordan curve $\partial\tilde{\Omega}$ with $\tilde{h}$ as its conformal welding.

Using the geodesic algorithm [19], we generate conformal maps $\tilde{\Phi}_1 : \mathbb{D} \to \tilde{\Omega}$ and $\tilde{\Phi}_2 : \mathbb{D}^c \to \tilde{\Omega}^c$ satisfying:

$$\tilde{\Omega} = \tilde{\Phi}_1(\mathbb{D}). \tag{14}$$

As the normalization conditions described in the equation 7, $\tilde{\Phi}_2(\infty) = \infty$ is required, then the $\tilde{\Omega}$ is ambiguous only up to translation and scaling. Without loss of generality, we can assume $\tilde{\Phi}_1(0) = 0$ and $\tilde{\Phi}_1(1) = 1$ to ensure the $\tilde{\Omega}$ is a uniquely defined.

4.4. Loss Function

The CW-VAE is trained by minimizing a composite loss function $\mathcal{L}_{\text{total}}$, which comprises three main components:

- KL divergence loss $\mathcal{L}_{KL}$ for latent space regularization
- Reconstruction loss $\mathcal{L}_{S\_recon}$ for welding vectors
- Auxiliary loss $\mathcal{L}_{Y\_recon}$ for conformal welding functions

Let S be the input welding vector and $\tilde{S}$ be the welding vector reconstructed by the CW-VAE. Let Y and $\tilde{Y}$ be the ground truth and reconstructed function value vectors, respectively. Let μ_z and σ_z^2 be the parameters of the learned latent Gaussian distribution $q_\phi(z|S)$. The prior distribution over latent variables is the standard normal distribution $p(z) = \mathcal{N}(0, I)$.

4.4.1. KL Divergence Loss ($\mathcal{L}_{KL}$) This standard VAE regularization term encourages the learned posterior distribution $q_\phi(z|S) = \mathcal{N}(z; \mu_z, \sigma_z^2)$ to approximate the prior $p(z)$. It is defined as:

$$\mathcal{L}_{KL} = D_{\text{KL}}(q_\phi(z|S) \| p(z)) = \frac{1}{2} \sum_{j=1}^{L_{\text{dim}}} (\mu_z^2 + \sigma_z^2 - \log(\sigma_z^2) - 1)_j. \tag{15}$$

where L_{dim} is the dimensionality of the latent space.

4.4.2. Feature Reconstruction Loss ($\mathcal{L}_{S\_recon}$) This loss quantifies the difference between the input welding vector S and the VAE's reconstructed welding vector $\tilde{S}$. We employ the Mean Squared Error (MSE):

$$\mathcal{L}_{S\_recon} = \frac{1}{N} \sum_{i=1}^{N} (s_i - \tilde{s}_i)^2. \tag{16}$$

This term ensures that the VAE accurately reconstructs the processed welding vector.

4.4.3. Conformal Welding Loss ($\mathcal{L}_{Y\_recon}$) To further guide the learning process and ensure that the generated outputs correspond to valid conformal welding functions, we introduce an auxiliary reconstruction loss directly on the function value vector Y. This loss is the MSE between the ground truth vector Y and the reconstructed vector $\tilde{Y}$ (which is derived from $\tilde{S}$ via the postprocessing steps described in Section 4.3):

$$\mathcal{L}_{Y\_recon} = \frac{1}{N+1}\sum_{i=0}^{N}(y_i - \tilde{y}_i)^2. \tag{17}$$

The inclusion of $\mathcal{L}_{Y\_recon}$ is crucial. Small, independent errors in the reconstructed welding vector $\tilde{S}$ can accumulate into large, undesirable global deviations in the function value vector $\tilde{Y}$, affecting its overall monotonic structure. This loss term encourages the VAE to learn representations that are not only precise in the S-space but also translate to globally well-behaved functions in the Y-space, promoting the generation of higher-quality and more geometrically plausible shapes.

4.4.4. Total Loss Function The overall loss function to be minimized is a weighted sum of these three components:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{S\_recon} + \lambda_Y \mathcal{L}_{Y\_recon} + \lambda_{KL}\mathcal{L}_{KL}. \tag{18}$$

where λ_Y and λ_{KL} are hyperparameters that control the relative importance of the conformal welding loss and the KL divergence term, respectively, compared to the primary feature reconstruction loss.

5. Experiments

To evaluate the effectiveness of our proposed CW-VAE framework, we conduct a series of experiments on diverse 2D shape datasets. We aim to assess the quality, diversity, and faithfulness of the generated shapes, and compare our method with relevant baselines. In this section, we describe the datasets used, the evaluation metrics, the experimental setup including implementation details, and the results obtained.

5.1. Datasets

We evaluate our method on two primary datasets, referred to as Dataset A and Dataset B, which represent different types of 2D shapes and challenges.

5.1.1. Dataset A: Animals Dataset A consists of 85 animal shapes collected from various online sources, representing 4 diverse animal categories, including birds, fish, dolphin and turtle. The shapes in this dataset are characterized by diverse morphologies with varying complexity. These shapes exhibit both smooth curves and sharp features that represent natural anatomical structures like tails, ears, and limbs. We preprocess these images of animals through the following steps: first segmenting the target animal shapes from their backgrounds, then extracting the boundary contours, followed by smoothing and resampling to obtain 200 evenly-spaced boundary points, and finally normalizing them to fit within a unit square while preserving their aspect ratios. Some segmentation masks are shown in Figure 5.

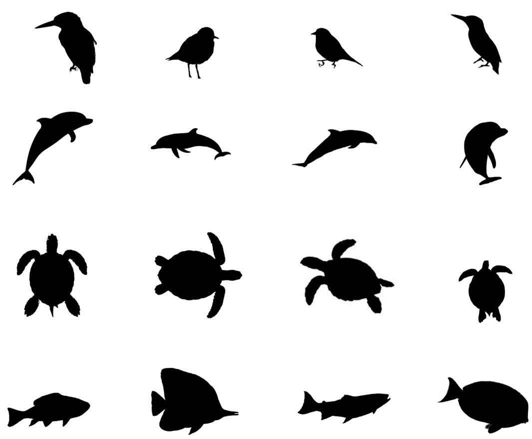


Figure 5: Some segmentation masks of animal shapes of 4 different categories in Dataset A.

5.1.2. Dataset B: X-ray Images of Lungs Dataset B comprises 63 lung contours extracted from chest X-ray images that we collected ourselves. This dataset is characterized by smoother boundaries with subtle anatomical variations that are clinically significant. The lung shapes exhibit both inter-subject variability and left-right asymmetry within subjects. We preprocess these lung X-ray images through the similar steps as in Dataset A except that we use the segmentation masks directly to extract the boundary contours. Worth noting that conformal welding is only applicable to simply-connected shapes, so we process left and right lungs separately. Some X-ray images and segmentation masks are shown in Figure 6.

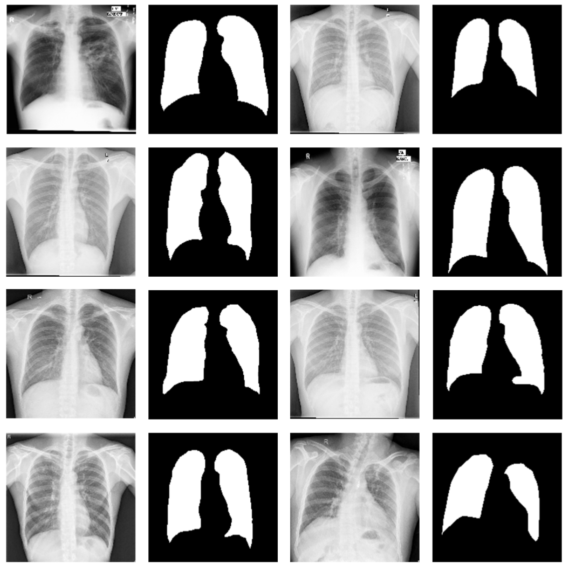


Figure 6: Some segmentation masks of lung shapes of 2 different categories in Dataset B.

5.2. Experimental Setup

Our CW-VAE model were implemented using Python 3.10 and PyTorch 2.5.1. All experiments were conducted on a server equipped with Intel Xeon Gold 6230 CPUs, NVIDIA RTX A6000 GPUs, and 1 TB of RAM, running Ubuntu 18.04 LTS.

5.2.1. CW-VAE Network Parameters For the CW-VAE architecture detailed in Section 4.2, the input welding vector S has a dimension of $N = 200$. The latent space dimension L_{dim} was set to 4 based on preliminary assessments balancing model capacity and complexity.

Both the encoder and decoder networks are constructed using a sequence of 3 ResLinearBlocks. The encoder maps the N-dimensional input vector to the L_{dim}-dimensional latent space parameters. The ResLinearBlocks in the encoder successively reduce dimensionality from $N \to 512 \to 256 \to 128$. The final 128-dimensional output is then projected to L_{dim} for both μ_z and $\log \sigma_z^2$ via separate linear layers.

Conversely, the decoder mirrors this structure, mapping the L_{dim}-dimensional latent sample z through ResLinearBlocks with expanding dimensions ($L_{\mathrm{dim}} \to$

$128 \to 256 \to 512$) before a final linear layer maps to the N-dimensional reconstructed vector $\tilde{S}$.

5.2.2. Training Parameters The CW-VAE model was trained using the Adam optimizer [15] with a constant learning rate of 5×10^{-5} and a weight decay of 1×10^{-5}. We used a batch size of 32 and trained the model for 5000 epochs. The loss function hyperparameters were set as $\lambda_Y = 1$ and $\lambda_{KL} = 0.5$.

5.2.3. Comparison Methods To demonstrate the efficacy of our proposed CW-VAE, we compared its performance against several baseline and alternative shape generation approaches. These include:

- **Standard VAE on Boundary Points:** A standard Variational Autoencoder trained directly on the raw coordinates of the uniformly resampled boundary points. This baseline helps to assess the benefit of using the conformal welding representation.
- **Image-based VAEs:** For qualitative comparisons, we compared against standard image-based generative models trained on binary rasterized images of the shapes. This highlights the differences between shape-intrinsic representations and pixel-based generation. The specific models used were Beta-VAE [14], a vanilla VAE [16], and a Spatial Broadcasting VAE [23].

For all methods, hyperparameters were tuned to ensure a fair comparison on the selected datasets.

5.3. Results and Analysis

5.3.1. Main Results In this section, we present results of our CW-VAE framework on animal datasets. The animal dataset contains 4 categories with approximately 20 images per category. We trained separate CW-VAE models for each animal category to evaluate the model's ability to learn and generate shapes within specific classes.

Figures 7 and 8 illustrate the generation results for birds and fish. The training data within each category exhibits substantial variation in terms of morphology, posture, angle, and size. Despite this high intra-class variability and the limited number of training samples, our CW-VAE successfully captures the characteristic features of each category.

The generated shapes demonstrate that our model can produce outputs that closely resemble the training data while also synthesizing novel variations, like the generated shape in row 2 column 3 of Figure 7 and the generated shape in row 4 column 2 of Figure 8.

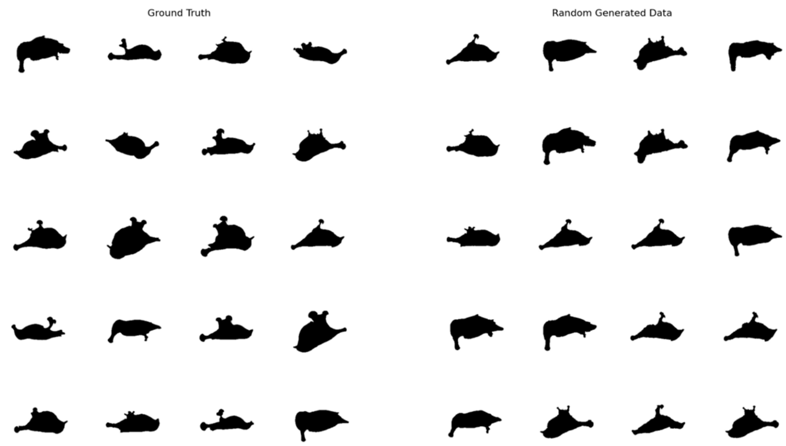


Figure 7: CW-VAE generation results for bird category. Left: examples from the training set. Right: shapes generated by sampling from the latent space.

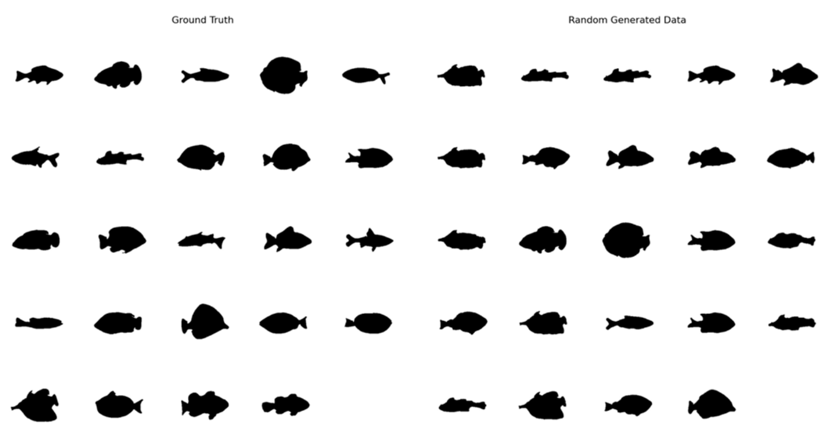


Figure 8: CW-VAE generation results for fish category. Left: examples from the training set. Right: shapes generated by sampling from the latent space.

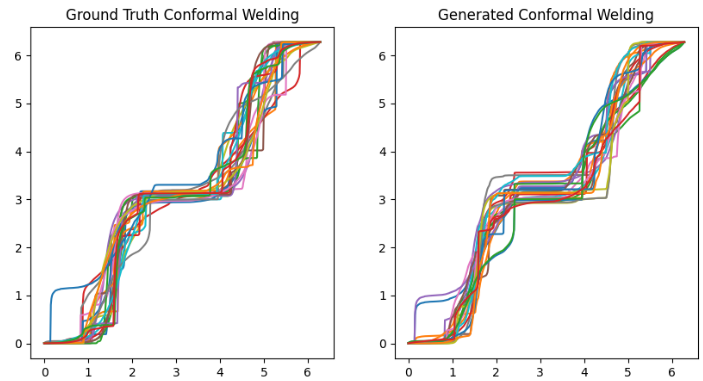


Figure 9: Comparison between original conformal welding functions $\{Y_i\}$ (left) and generated welding functions $\{\tilde{Y}_j\}$ (right) for fish category.

To provide deeper insight into how CW-VAE captures shape characteristics, we visualize the conformal welding functions $\{Y_i\}$ and the generated $\{\tilde{Y}_j\}$ for the fish category in Figure 9. Each curve in this visualization corresponds to a specific shape. The generated welding functions $\{\tilde{Y}_j\}$ (shown in right) exhibit similar properties to the original training data welding functions $\{Y_i\}$ (shown in left), such as the characteristic flatter slope in the middle section of curves. The model also successfully preserves patterns for distinctive curves, like the sharp increase in the beginning, demonstrating that CW-VAE effectively learns the intrinsic features of the conformal welding representation.

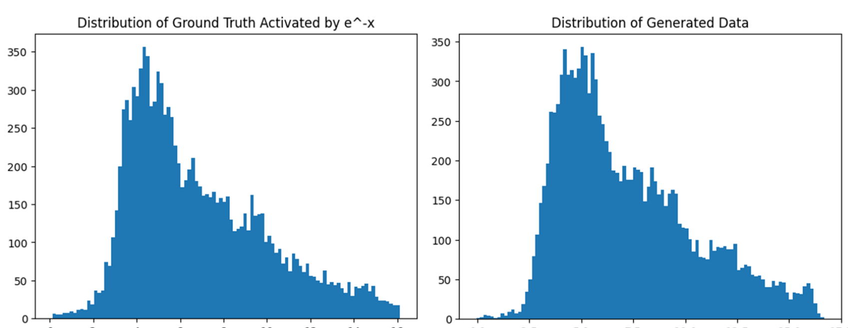


Figure 10: Distribution of welding vectors for fish category. Left: original training data $\{S_i\}$. Right: generated welding vectors $\{\tilde{S}_j\}$.

The distribution of welding vectors in Figure 10 further confirms this observation, showing that the model has captured the underlying statistical properties of the welding vector representation space. This suggests that the conformal welding approach with our designed pre- and post-processing provides a meaningful and learnable shape prior that can be effectively utilized by deep generative models.

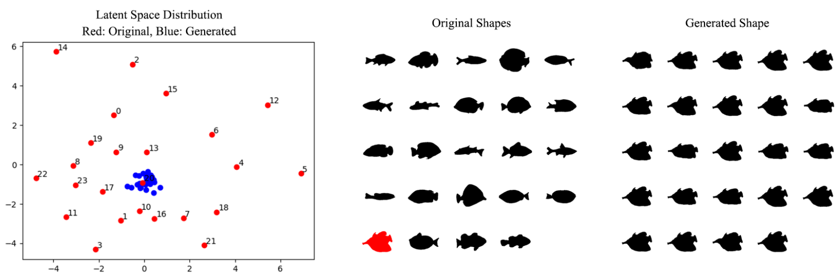


Figure 11: Latent space exploration and shape sampling. Left: Distribution of training shapes in the latent space, with each red point representing a training shape labeled by its index in the ground truth panel (center) and each blue points representing sampling points taken around shape #20 to generate new shapes. Center: Ground truth shapes with shape #20 highlighted in red. Right: Generated shapes corresponding to the blue sampling points in the latent space (left).

To further investigate the structure of the learned latent space, we visualize the distribution of training data in Figure 11. Although the latent space dimension $L_{\text{dim}} = 4$, for visualization purposes, we project the latent codes onto their first two principal components. Each red point in the left panel represents a training shape, with its corresponding index matching the shape's position in the center panel. The latent space organization demonstrates that the CW-VAE has learned a meaningful embedding where similar shapes are positioned closer together.

This structured latent space allows us to sample new shapes in a controlled manner. To demonstrate this capability, we sample around shape #20 (highlighted in red in the center panel). The blue points in the left panel represent these latent samples, and their corresponding generated shapes are shown in the right panel. These generated shapes exhibit strong similarity to the original reference shape while displaying subtle variations in their contours. Some generated instances show more pronounced yet plausible differences, such as flatter body shapes, representing novel variations that maintain the essential characteristics of the original. This demonstrates that our CW-VAE not only learns to reconstruct existing shapes but also enables controlled generation by navigating the latent space.

5.3.2. Application on Lung Contour Dataset To further validate the effectiveness of our CW-VAE framework, we applied it to the medical imaging data, specifically the lung contour dataset described in Section 5.1. Medical imaging presents a particularly suitable application domain for our approach due to several key characteristics: limited data availability, the presence of objects with significant geometric features, and high requirements for interpretability. These conditions align well with the strengths of our CW-VAE framework.

Since conformal welding is applicable only to simply-connected shapes, we trained separate models for left and right lung contours. This separation is anatomically justified, as left and right lungs exhibit notable structural differences in shape and size due to the asymmetric positioning of organs in the thoracic cavity.

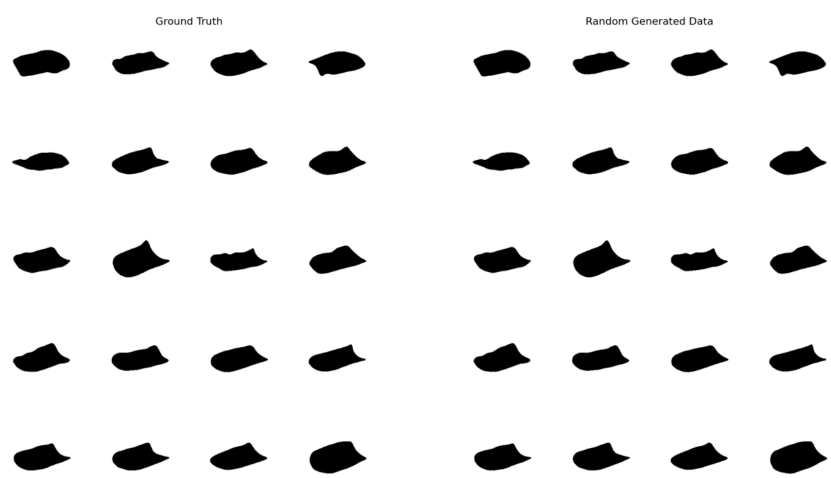


Figure 12: CW-VAE generation results for left lung contours. Left: examples from the training set. Right: shapes generated by sampling from the latent space.

Figures 12 and 13 present the generation results for left and right lung contours, respectively. Similar to our experiments with animal shapes, the

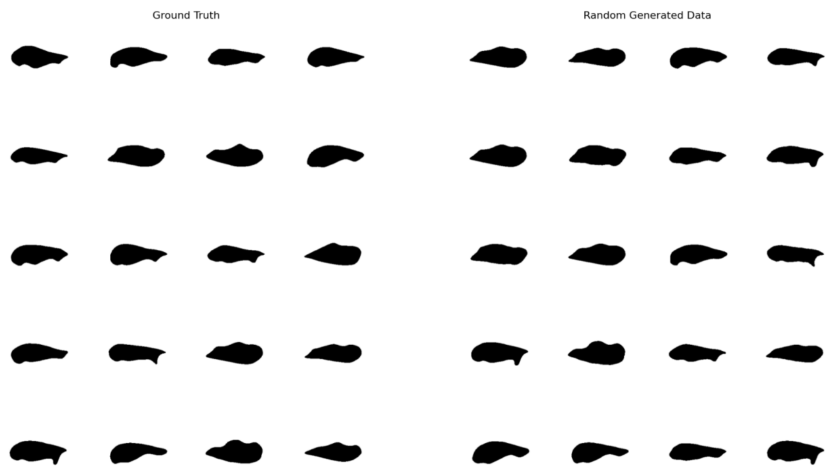


Figure 13: CW-VAE generation results for right lung contours. Left: examples from the training set. Right: shapes generated by sampling from the latent space.

CW-VAE demonstrates strong performance in capturing the essential geometric characteristics of lung contours. The generated shapes maintain the anatomical plausibility expected in medical applications while exhibiting natural variations. This capability could be valuable for data augmentation in downstream tasks, such as training segmentation algorithms or conducting anatomical variability studies.

The successful application of CW-VAE to lung contours highlights its potential utility in medical imaging contexts where data scarcity is a common challenge. By effectively learning the shape prior from a limited set of examples, our approach offers a promising solution for generating realistic anatomical shapes that can support various medical image analysis tasks.

5.3.3. Comparison with Direct Boundary Point Generation While our CW-VAE framework has demonstrated promising results, a natural question arises: Is it necessary to transform boundary points into conformal welding representations before learning, or could we directly learn from the boundary points themselves? To address this question, we conducted a comparative analysis between our CW-VAE and a baseline VAE that learns directly from the boundary points themselves.

For the point-based VAE, we used the concatenated x and y coordinates of the boundary points as the training data and fed them into a VAE with an architecture identical to the backbone of our CW-VAE. All other experimental settings, including latent space dimension and training parameters, were kept identical to ensure a fair comparison.

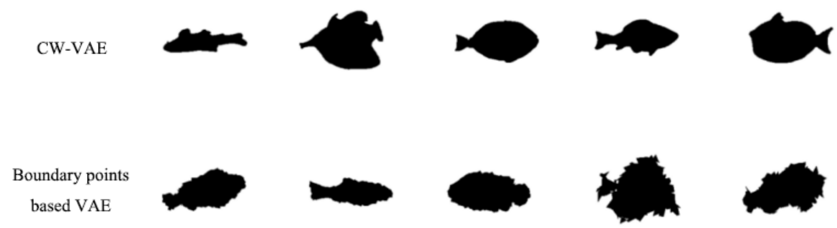


Figure 14: Comparison between CW-VAE (top) and boundary points-based VAE (bottom). The boundary points-based approach produces shapes with noticeable jaggedness compared to our CW-VAE approach.

As shown in Figure 14, the performance of the boundary points-based VAE is noticeably inferior to that of our CW-VAE. The shapes generated by the boundary points-based approach exhibit significant jaggedness along their boundaries and demonstrate less diversity, with many generated samples appearing highly similar to each other. These limitations can be attributed to several key factors:

1. **Representation Complexity:** Boundary point coordinates form a 2D sequence, making them inherently more complex to model compared to the 1D representation of a conformal welding function. This increased complexity makes the learning task more challenging for the VAE.
2. **Geometric Feature Encoding:** The normalized conformal welding function is a more direct and disentangled representation of a shape's intrinsic geometry. Similarity between conformal welding functions often better reflects the perceptual similarity between their corresponding shapes, making the representation space more structured and easier to learn.
3. **Learning Difficulty:** For a VAE, generating a smooth, closed, and non-self-intersecting sequence of boundary points is a difficult task. In contrast, our approach offloads the geometric heavy lifting to the deterministic conformal welding computation and reconstruction algorithms. This allows the VAE to focus on learning the high-level distribution of shapes rather than struggling with low-level geometric constraints.

These results empirically validate the value of incorporating strong, mathematically-grounded shape priors into generative models. The conformal welding representation corresponds to a more clearly structured and learnable space, leading to higher-quality and more diverse shape generation.

5.3.4. Comparison with Pixel-based Generation To further validate the advantages of our shape-prior-based approach, we compared CW-VAE with several pixel-based generative models that operate directly on binary image representations of shapes. Most conventional VAE approaches generate images by modeling pixel-level similarities, often without explicitly considering the geometric properties of the shapes contained within. We conducted comparisons with three representative pixel-based VAE variants: Beta-VAE [14], a vanilla VAE [16], and a Spatial Broadcasting VAE [23].

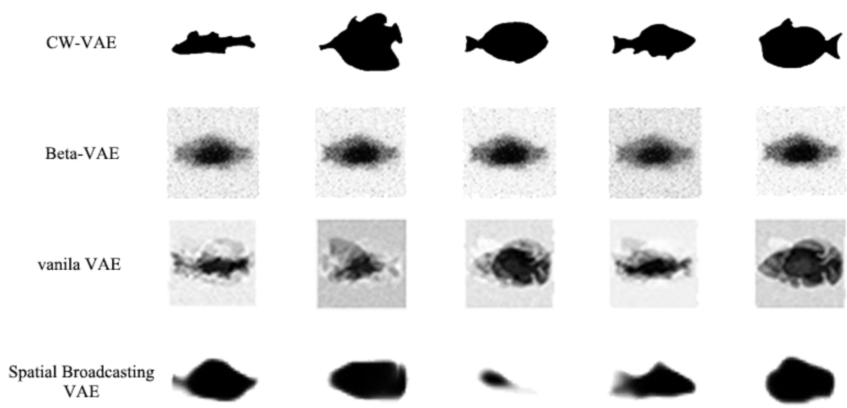


Figure 15: Comparison between our CW-VAE and pixel-based VAE approaches. From top to bottom: CW-VAE, Beta-VAE, vanilla VAE, and Spatial Broadcasting VAE.

As shown in Figure 15, all three pixel-based VAE variants significantly underperform compared to our CW-VAE. Each approach exhibits distinct limitations when applied to shape generation with limited training data:

1. **Beta-VAE:** This approach produces extremely blurry images with minimal variance between samples, a clear sign of mode collapse or posterior collapse. By increasing the KL divergence penalty via the β parameter, Beta-VAE forces a more disentangled latent space. However, with only about 20 training samples, the model cannot learn sufficient distributional characteristics and instead collapses all inputs to a single point

in the latent space, causing the decoder to generate nearly identical outputs.

2. **Vanilla VAE:** While faint outlines are visible, the generated shapes exhibit significant blurriness and noise artifacts. The continuous Gaussian latent space of a standard VAE struggles to effectively represent the inherently sharp and structured nature of binary shape boundaries. With limited data, the model fails to properly separate shape features from background noise, resulting in a decoder that relies on averaging pixel values rather than learning meaningful shape features.
3. **Spatial Broadcasting VAE:** This approach produces blocky, blurred regions that fail to capture local geometric details. Its broadcasting mechanism, which expands low-dimensional latent variables across a spatial grid, is ill-suited for learning fine-grained boundary information from a small dataset. With insufficient training examples, the model cannot learn the local geometric constraints of shape contours, such as curves or corners, leading to outputs that lack structural coherence.

These results highlight the fundamental advantages of our CW-VAE approach:

- **Dimensionality Reduction:** Conformal welding transforms a high-dimensional pixel representation into a compact 1D function, which is naturally better suited for small-sample scenarios.
- **Explicit Geometric Constraints:** By leveraging a prior from complex analysis, our approach ensures that generated shapes maintain their geometric integrity without blurriness or structural distortion.
- **Decoupled Generation Process:** Our two-stage approach (feature generation followed by deterministic reconstruction) allows the VAE to focus solely on learning the distribution of conformal welding representations rather than the complex image rendering process, significantly reducing the learning difficulty.

The superior performance of CW-VAE in this comparison further validates our core thesis that incorporating appropriate geometric priors into generative models can substantially improve shape generation quality, especially in scenarios with limited training data.

6. Conclusion and Future Work

This paper presents the Conformal Welding Variational Autoencoder (CW-VAE), a novel framework for 2D shape generation that integrates conformal welding as a geometric prior. Our approach effectively addresses the challenge of generating diverse, high-quality shapes from limited training data. The key contributions include:

- A complete shape generation pipeline that transforms boundary points into conformal welding representations, learns their distribution via VAE, and reconstructs new shapes
- Preprocessing techniques that normalize conformal weldings for uniqueness and transform derivative vectors for better VAE learning
- A specialized VAE architecture with residual connections and a composite loss function ensuring geometric validity

Our experimental results demonstrate CW-VAE's clear advantages over alternative approaches. Unlike boundary points-based VAEs that produce jagged contours, or pixel-based approaches that suffer from mode collapse, blurriness, and loss of structural coherence in small-data regimes, CW-VAE generates diverse, geometrically valid shapes even with highly limited training data.

The success of CW-VAE stems from three key design principles: (1) dimensionality reduction through the conformal welding representation, requiring significantly fewer parameters than pixel representations; (2) incorporation of explicit geometric constraints derived from complex analysis; and (3) a decoupled generation process that separates representation learning from shape reconstruction. These findings validate our core idea that incorporating appropriate geometric priors into generative models substantially improves shape generation quality in data-scarce scenarios.

There are several promising directions for future work:

- **Extension to More Complex Topologies and Dimensions:** The current approach is limited to simply-connected 2D shapes. Future work could explore extensions to multiply-connected domains or adapt the framework to 3D shapes, perhaps by using panoramic conformal parameterizations of genus-zero surfaces.
- **Integration with Advanced Generative Frameworks:** While we focused on a VAE, the conformal welding representation could be integrated into other generative frameworks like Diffusion Models or Normalizing Flows, potentially yielding even higher-quality results.
- **Conditional Generation and Shape Editing:** Developing methods for conditional shape generation based on attributes or classes would enhance practical utility. The structured latent space of CW-VAE also presents a promising foundation for intuitive shape editing and interpolation tools.

- **Exploration of Alternative Shape Priors:** While conformal welding is a powerful prior, investigating other geometric representations, such as the Harmonic Beltrami Signature (HBS), could yield complementary benefits or address different classes of shapes and applications.

In conclusion, our work demonstrates that the synergy between classical geometry and deep generative models provides a robust path to overcoming the data limitations that currently constrain shape modeling. CW-VAE opens new possibilities for shape analysis and synthesis across domains from computer graphics to medical imaging.

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Chenran Lin
Department of Mathematics, The Chinese University of Hong Kong, Hong Kong
E-mail address: crlin@math.cuhk.edu.hk

Oscar Yau Lam Chau
Department of Mathematics, The Chinese University of Hong Kong, Hong Kong
E-mail address: 1155174316@link.cuhk.edu.hk

Lok Ming Lui
Department of Mathematics, The Chinese University of Hong Kong, Hong Kong
E-mail address: lmlui@math.cuhk.edu.hk